

Inferential Statistics: Knowledge Score vs. Age Group

Independent Two-Sample T-Test

This report examines whether there is a statistically significant difference in clinical knowledge between carriers who are younger than 35 and those who are 35 or older.

Descriptive Statistics

Age Group	N	Mean (V3 Score)	Std Dev (V3 Score)	Mean (Weighted V3)	Std Dev (Weighted)
Below 35	149	11.74	7.94	2.164	4.431
35 and Above	50	9.56	8.27	1.013	4.566

Hypothesis Testing (Welch's T-Test)

We use Welch's T-Test (assuming unequal variances) for robust results.

1. Analysis using the mathematically rigorous Weighted V3 Knowledge Score:

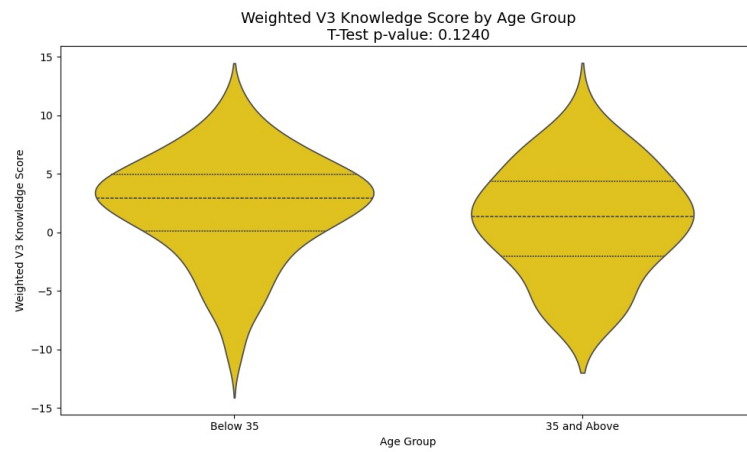
- **T-Statistic:** -1.554
- **P-Value:** 0.1240
- **Conclusion:** Not Statistically Significant ($p > 0.05$). There is no significant difference in knowledge based on this age split.

2. Analysis using the standard V3 Knowledge Score:

- **T-Statistic:** -1.627
- **P-Value:** 0.1076
- **Conclusion:** Not Statistically Significant ($p > 0.05$).

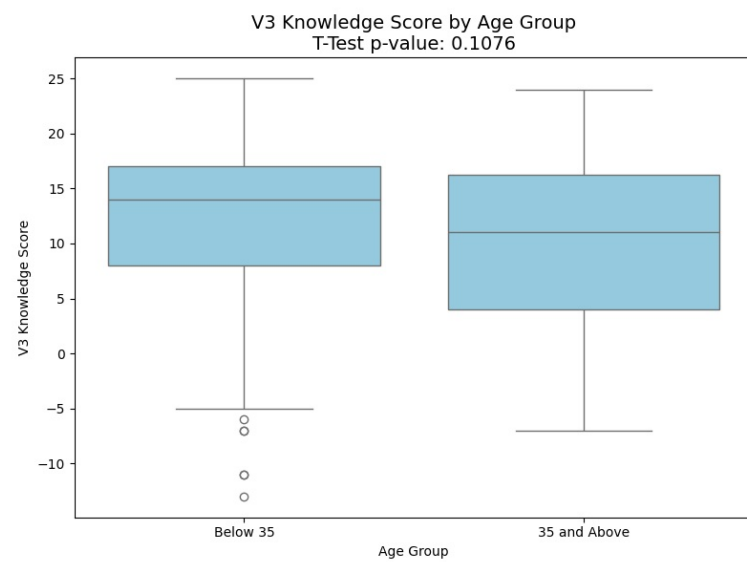
Visualizations

Weighted V3 Score Violin Plot



Violin Plot

Standard V3 Score Box Plot



Box Plot